

(1001CJA621120025)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

10-01-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	D	A	B	A	A,C	A,D	B,C
SECTION-II	Q.	11	12								
	A.	B,C	A,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	2.00	6.00	1.00	24.00	4.00	2.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	C	B	B	D	B,C	A,B,C	A,B,D	B,C
SECTION-II	Q.	11	12								
	A.	A,B,D	A,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.75	2.00	4.50	9.00	3.00	6.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	D	D	D	B	B,C,D	C,D	A,C	A,B,C
SECTION-II	Q.	11	12								
	A.	A,B	A,B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	4.00	2.00	2.00	4.00	2.00	7.00				

(1001CJA621120026)

Test Pattern



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JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	
	A.	A,D	B,C,D	B,C	A,C	A,D	A,D	
SECTION-II	Q.	1	2	3	4	5	6	
	A.	12.00	1800.00	148.00	3.09	1.00	42.00	
SECTION-III	Q.	1	2	3	4	5	6	
	A.	8	3	4	6	1	2	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	
	A.	A,B,C	A,B,D	B,C	A,D	A,B,D	A,C,D	
SECTION-II	Q.	1	2	3	4	5	6	
	A.	8.90 - 9.00	2.00	22.00	28.00	10.00	50.00	
SECTION-III	Q.	1	2	3	4	5	6	
	A.	6	8	3	7	3	3	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	
	A.	A,C,D	A,D	B,C	B,C,D	A,B,C	A,C	
SECTION-II	Q.	1	2	3	4	5	6	
	A.	500.00	13.00	0.06	4.00	0.50	8.00	
SECTION-III	Q.	1	2	3	4	5	6	
	A.	1	1	3	4	1	2	

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)**PAPER-1****PART-2 : CHEMISTRY****SOLUTION****SECTION-I**1. **Ans.(A)**

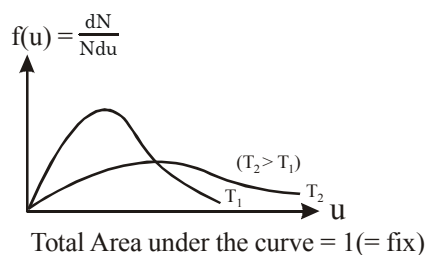
B,D : ideal solution

C : Negative deviation

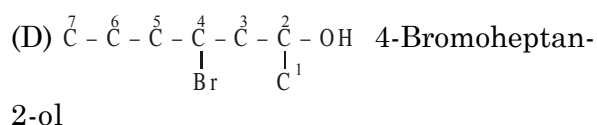
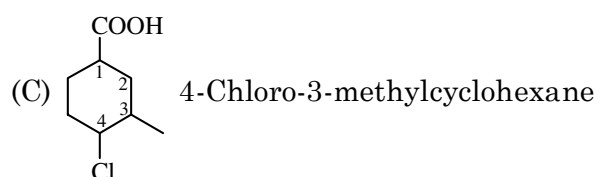
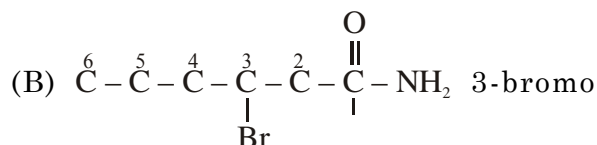
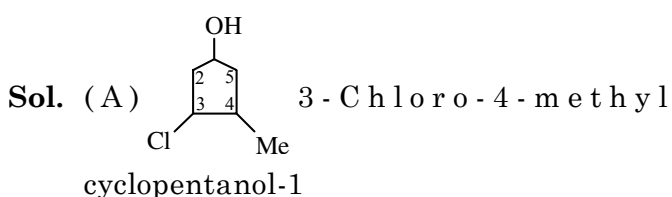
2. **Ans.(B)**

$$k = \frac{1}{25} \ln \left(\frac{42-18}{42-30} \right) = \frac{\ln 2}{25}$$

$$t_{1/2} = \frac{\ln 2}{K} = 25$$

3. **Ans.(C)**4. **Ans.(B)**5. **Ans.(B)**6. **Ans.(D)**7. **Ans.(B,C)**

$$u_{mp} \propto \sqrt{T}$$

8. **Ans.(A,B,C)**9. **Ans.(A,B,D)**10. **Ans.(B,C)**11. **(A,B,D)**12. **Ans. (A,C)****SECTION-II**1. **Ans.(0.75)**

$$Z = \frac{PM}{\rho RT} = \frac{10^5 \times (18 \times 10^3)}{0.6 \times 8 \times 500} = 0.75$$

2. **Ans.(2.00)**

$$P = K_H \cdot X \Rightarrow 1.6 \times 10^7 = 80 \times 10^3 \times 10^5 \times$$

$$\frac{n_{N_2}}{18 \times 10^3 / 18} \Rightarrow n_{N_2} = 2.$$

3. **Ans.(4.50)**4. **Ans.(9.00)**5. **Ans. (3.00)**6. **Ans. (6.00)****(ii), (iii), (iv), (vi), (vii), (ix)**

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Put $x = 2 + h$

$$L = \lim_{h \rightarrow 0} \frac{[h]}{\operatorname{sgn}(h)} = \frac{0}{1} = 0$$

Put $x = 2 - h$

$$M = \lim_{h \rightarrow 0} \frac{[-h]}{\operatorname{sgn}(-h)} = \frac{-1}{-1} = 1$$

$$L + M = 1$$

2. **Ans. (A)**

$$\text{LHL} = \lim_{x \rightarrow 0^-} a \cot^2 x \cdot \ln \cos x$$

$$= \frac{a \ln(1 + (\cos x - 1))}{\tan^2 x} = \frac{a(\cos x - 1)}{x^2} = -\frac{a}{2}$$

$$f(0) = 1$$

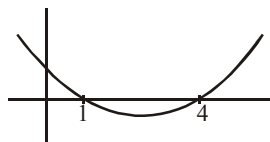
$$\text{RHL} = \lim_{x \rightarrow 0^+} (\sec x)^{b \cot x}$$

$$= e^{(\sec x - 1)b \frac{\cos x}{\sin x}} = e^{\frac{b(1 - \cos x) \cdot x}{x^2}} = e^0 = 1$$

$$\Rightarrow a = -2 \text{ \& } b \in \mathbb{R}$$

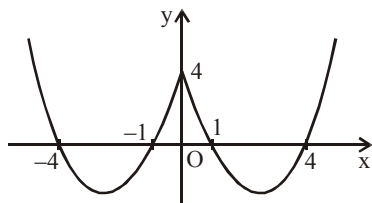
3. **Ans. (D)**

Let us plot $f(x) = x^2 - 5x + 4$

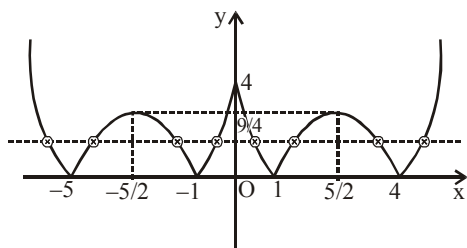


Now $x \rightarrow |x|$

$\Rightarrow y = f(|x|)$ will be



Now $f(|x|) \rightarrow |f(x)|$



From the above graph it is clear that number of points of intersection of $y = |f(x)|$ & $y = 2$ is

4. **Ans. (D)**

When at even positions OI

$${}^6C_2 \times {}^6C_3 \times 2! \times 3! \times \frac{7!}{2!} = 1800 \times 7!$$

When at even positions AI or AO

$${}^6C_2 \times {}^6C_3 \times 2 \times 2 \times \frac{3!}{2!} \times \frac{7!}{2!} = 1800 \times 7!$$

$$\therefore \text{Total arrangement} = 3600 \times 7!$$

5. **Ans. (D)**

$$S_1P + S_2P = 2a \text{ \& } S_1P - S_2P = 2A$$

$$S_1P = a + A \text{ \& } S_2P = (a - A)$$

$$\text{quadratic equation is } x^2 - 2ax + (a^2 - A^2) = 0$$

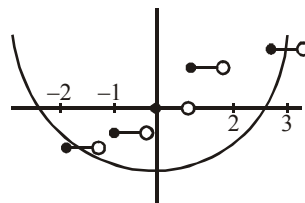
6. **Ans. (B)**

$$-1 \leq [x] - x^2 + 4 \geq 2$$

$$-5 \leq [x] - x^2 \leq -2$$

$$(i) -5 \leq [x] - x^2 \Rightarrow x^2 - 5 \leq [x]$$

Intersection for both the curves above x axis lies between (2,3)



$$x^2 - 5 = 2 \Rightarrow x = \sqrt{7}$$

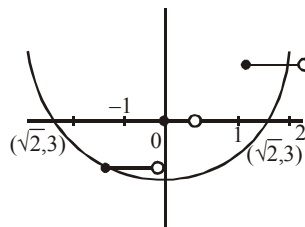
below x axis between (-2, -1)

$$x^2 - 5 = -2 \Rightarrow x = -\sqrt{3}$$

$$x = -\sqrt{3}$$

$$(ii) [x] - x^2 \leq -2 \Rightarrow x^2 - 2 \geq [x]$$

above x-axis between (1,2)



$$x^2 - 2 = 1 \Rightarrow x = \sqrt{3}$$

below axis it is at $x = -1$

$$\text{The interval is } (-\infty, -1] \cup (\sqrt{3}, \infty)$$

Intersection of case (i) & (ii) gives option (B)

7. Ans. (B, C, D)

- (a) $(f(x) + g(x)) - h(x) = \text{cont.} - \text{discont.}$
will always be discontinuous
(b) $(f(x)g(x)) h(x) = (\text{cont}) (\text{discont})$,
may or may not be cont
(c) True
(d) $\frac{\text{cont.}}{\text{discont}}$, may or may not be continuous

8. Ans. (C, D)

$$f(f(x)) = \frac{2f(x)+1}{f(x)} = \frac{\frac{2(2x+1)}{x}+1}{\frac{2x+1}{x}} = \frac{5x+2}{2x+1}$$

$$g(x) = f(f(x)) - \frac{1}{f(x)}, x \neq 0, -\frac{1}{2}$$

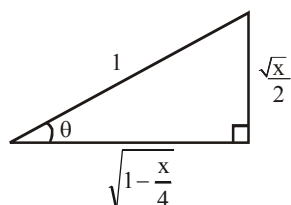
$$g(x) = \frac{5x+2}{2x+1} - \frac{x}{2x+1}$$

$$g(x) = 2$$

$\Rightarrow g(x)$ is a constant function

9. Ans. (A, C)

$$\sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right) = \theta \Rightarrow \sin \theta = \sqrt{1-\frac{x}{4}}$$



$$\frac{\pi}{2} + \tan^{-1} y = \frac{2\pi}{3} \Rightarrow y = \frac{1}{\sqrt{3}}$$

$$\Rightarrow 0 \leq x \leq 4$$

$$\text{Max. value of } x^2 + y^2 \text{ is } \frac{49}{3}$$

$$\text{Min. value of } x^2 + y^2 \text{ is } \frac{1}{3}$$

10. Ans. (A, B, C)

$$(A) \text{ Tangent of slope } m \text{ is } y = mx + \frac{3}{m}$$

$$\text{For } m = 3, \text{ tangent is } y = 3x + 1$$

$$(B) \text{ Normal of slope } m \text{ is}$$

$$y = mx - 6m - 3m^3$$

$$\text{For } m = 2, \text{ normal is } y = 2x - 36$$

(C) Directrix is $x = -3$

Any point on it is $(-3, k)$

Its chord of contact is $ky = 6(x - 3)$

$$\text{or } 6x - ky = 18 \quad \text{or } 2x - \frac{k}{3}y = 6$$

(D) Line $x - 2y + 12 = 0$ is

a tangent to $y^2 = 12x$

at $(12, 12)$

11. Ans. (A, B)

$$\frac{x}{ma} + \frac{y}{nb} = 1 \Rightarrow y = \frac{-nb}{ma}x + nb \quad \dots(1)$$

but any tangent to the ellipse is

$$y = Mx \pm \sqrt{a^2M^2 + b^2} \quad \dots(2)$$

comparing (1) & (2)

$$M = -\frac{nb}{ma}, a^2M^2 + b^2 = n^2b^2$$

$$a^2 \frac{n^2b^2}{m^2a^2} + b^2 = n^2b^2$$

$$\frac{n^2}{m^2} + 1 = n^2$$

$$\Rightarrow m^2 = \frac{n^2}{n^2 - 1}, n^2 = \frac{m^2}{m^2 - 1}$$

12. Ans. (A, B, C, D)

$$(1)^\infty \text{ form; use } \lim_{x \rightarrow 0} e^{\frac{4}{5x^2}(a \sin bx^2 + 4(1 - \cos^2 x))} = e^2$$

$$\Rightarrow a \frac{\sin bx^2}{x^2} \times \frac{4}{5} + \frac{16 \sin^2 x}{5x^2} = 2$$

$$\frac{4ab}{5} + \frac{16}{5} = 2 \Rightarrow 4ab = -6 \Rightarrow ab = -\frac{2}{3}$$

SECTION-II

1. Ans. 4.00

$$f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$$

$$-1 \leq [2x] \leq 1 \quad \& \quad -1 \leq [x] - 1 \leq 1$$

$$\Rightarrow -1 \leq 2x < 2 \quad \& \quad 0 \leq [x] \leq 2$$

$$\Rightarrow -\frac{1}{2} \leq x < 1 \quad \& \quad 0 \leq x < 3$$

$$\Rightarrow 0 \leq x < 1 \text{ Domain}$$

$$\Rightarrow [x] = 0 \Rightarrow 0 \leq 2x < 2$$

$$\Rightarrow [2x] = 0 \text{ or } 1$$

$$\text{Now } f(x) = \sin^{-1}[2x] + \cos^{-1}(-1)$$

$$= \left(0 \text{ or } \frac{\pi}{2}\right) + \pi = \pi \text{ or } \frac{3\pi}{2}$$

$$\Rightarrow a + b + \frac{2d}{c} = 1 + 3 = 4$$

2. **Ans. 2.00**

$$\text{Let } \operatorname{cosec}^{-1}x = \theta \Rightarrow \operatorname{cosec}\theta = \frac{x}{1}$$

$$\sec\theta = \frac{x}{\sqrt{x^2 - 1}}$$

$$f(x) = \frac{x}{\sqrt{x^2 - 1}}$$

$$\lim_{x \rightarrow 2} \frac{\frac{x}{\sqrt{x^2 - 1}} - \sqrt{3}}{(x - 2)} = \lim_{x \rightarrow 2} \frac{\sqrt{x^2 - 1} - \sqrt{3}}{(x - 2)}$$

$$\lim_{x \rightarrow 2} \frac{(x^2 - 4)}{(x - 2)(\sqrt{x^2 - 1} + \sqrt{3})} = \frac{4}{2\sqrt{3}} = \frac{2}{\sqrt{3}}$$

3. **Ans. 2.00**

$$2(g_1g_2 + f_1f_2) = c_1 + c_2$$

$$\Rightarrow 2\left(n_1\left(\frac{n_2}{2}\right) + (1)\left(\frac{n_2}{2}\right)\right) = n_1$$

$$\Rightarrow n_1n_2 + n_2 = n_1$$

$$\Rightarrow n_2 = \frac{n_1}{(1 + n_1)} \Rightarrow n_2 = 1 - \frac{1}{(1 + n_1)}$$

$$1 + n_1 = 1, 1 + n_1 = -1$$

$$n_1 = 0 \& n_1 = -2 \Rightarrow n_2 = 0 \& n_2 = 2$$

4. **Ans. 4.00**

${}^4C_0 \cdot {}^{404}C_4 - {}^4C_1 \cdot {}^{303}C_4 + {}^4C_2 \cdot {}^{202}C_4 - {}^4C_3 \cdot {}^{101}C_4$
the given expression is the coefficient of x^4 in

$${}^4C_0(1 + x)^{404} - {}^4C_1(1 + x)^{303} + {}^4C_2(1 + x)^{202} - {}^4C_3(1 + x)^{101} + {}^4C_4$$

$$\text{i.e. in } ((1 + x)^{101} - 1)^4$$

$$\Rightarrow ({}^{101}C_1x + {}^{100}C_2x^2 + \dots + {}^{101}C_{101}x^{101})^4$$

$$\Rightarrow ({}^{101}C_1)^4 = 101^4$$

$$(101)^4 = (101^2)^2 = (10201)^2$$

$$a = 10201$$

$$\text{sum of digits} = 4$$

5. **Ans. 2.00**

$$P(x) = 3x^3 - 9x^2 + 2x + 5$$

$$P'(x) = 9x^2 - 18x + 2$$

$$P''(x) = 18x - 18$$

$$\Rightarrow f(x) = \begin{cases} 3x, & x \neq \frac{\pi}{2} \\ \sin^{-1}(ab) + \cos^{-1}(a + b - 3ab), & x = \frac{\pi}{2} \end{cases}$$

For continuity

$$\sin^{-1}(ab) + \cos^{-1}(a + b - 3ab) = \frac{3\pi}{2}$$

$$\Rightarrow ab = 1 \& a + b - 3ab = -1$$

$$\Rightarrow a = b = 1 \Rightarrow a + b = 2$$

6. **Ans. 7.00**

$$T_n = \tan^{-1} \frac{2}{n^2} = \tan^{-1} \left[\frac{(n+1) - (n-1)}{1 + (n-1)(n+1)} \right]$$

$$= \tan^{-1}(n+1) - \tan^{-1}(n-1)$$

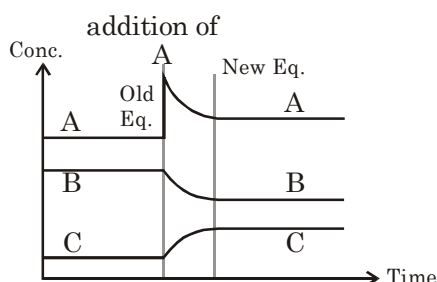
$$\sum_{n=1}^n \tan^{-1} \frac{2}{n^2} = \tan^{-1}18 + \tan^{-1}17 - \tan^{-1}1$$

$$= \pi - \tan^{-1} \frac{34}{27}$$

$$p - q = 7$$

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)**PAPER-2****PART-2 : CHEMISTRY****SOLUTION****SECTION-I**1. **Ans.(A,B,C)**

K_p = Function of temperature only for a given equilibrium

2. **Ans.(A,B,D)**

(A) On increasing temperature more electron goes to conduction band.

(C) MnO is anti ferromagnetic solid.

3. **Ans.(B,C)**4. **Ans.(A,D)**5. **Ans.(A, B, D)**6. **Ans.(A,C,D)****SECTION-II**1. **Ans.(8.90 - 9.00)**

$$4 = 2 \times 1.8 \times \frac{32/(120 + 18x)}{84 + \left[\frac{32}{120 + 18x} \times 18x \right]}$$

$$1000$$

$$4 = 3.6 \times \frac{32000}{84(120 + 18x) + 32 \times 18x}$$

$$x = 8.96$$

2. **Ans.(2.00)**

Sol. After 1st half life number of atoms

$$= 5 \times 10^{23} + 10^{23}$$

After 2nd half life number of atoms

$$= 3 \times 10^{23} + 10^{23}$$

Number of atoms disintegrated between 2nd and 3rd half life = 2×10^{23}

3. **Ans.(22.00)**Sol. $x = 4$; $y = 2$;

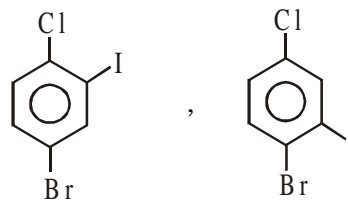
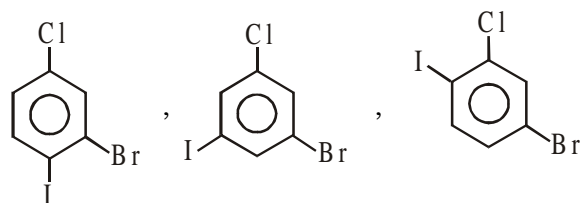
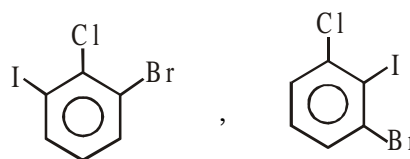
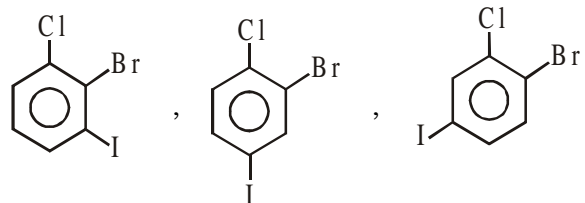
$$z = 3 \times 5 + 1 = 16$$

$$x + y + z = 4 + 2 + 16 = 22$$

4. **Ans.(28.00)**Sol. $x = B_2H_6, Si_2H_6, Al_2Cl_6, BH_3 \cdot NH_3$ (4)

$y = B_2H_6, Be_2H_4, Si_2H_6, Al_2Cl_6, I_2Cl_6, BH_3 \cdot NH_3, \cdot CH_3$ (7)

$$x \times y = 4 \times 7 = 28$$

5. **Ans. (10.00)**6. **Ans. (50.00)**

SECTION-III

1. Ans.(6)

$$X \rightarrow Y + \alpha$$

$$t = 0 \quad 0.8$$

$$t = t \quad 0.8 - x \quad x = \frac{15.68}{22.4} = 0.7$$

$$\Rightarrow t = 3t_{1/2} = 6 \text{ hrs}$$

2. Ans.(8)

$$\log \frac{k_2}{k_1} = \frac{9200}{2.3 \times 2} \left[\frac{1}{500} - \frac{1}{1000} \right]$$

$$\frac{ROR_2}{ROR_1} = \frac{k_2}{k_1} = 100$$

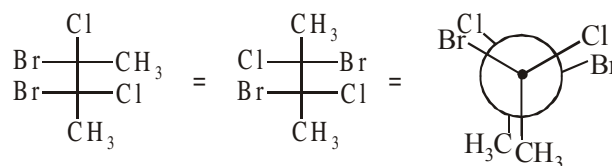
$$\Rightarrow ROR_2 = 8 \text{ mol L}^{-1} \text{ min}^{-1}$$

3. Ans.(3)

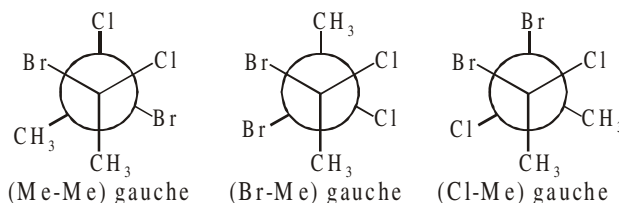
Sol. ($\text{NF}_3 < \text{N}(\text{Me})_3$) ; ($\text{SiCl}_4, \text{CCl}_4$), (H_2, O_2)

4. Ans.(7)

5. Ans.(3)



Stable conformer (with $\mu \neq 0$)



6. Ans.(3)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans.(A,C,D)

$$L = \lim_{x \rightarrow 0} \frac{a \left(x - \frac{x^3}{6} + \frac{x^5}{120} \dots \right) - bx + cx^2 + x^3}{2x^2 \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \right) - 2x^3 + x^4}$$

$$= \lim_{x \rightarrow 0} \frac{(a-b)x + cx^2 + \left(1 - \frac{a}{6}\right)x^3 + \frac{ax^5}{120}}{\frac{2x^5}{3} - \frac{x^6}{2} + \dots}$$

$$\Rightarrow a = b, c = 0, 1 - \frac{a}{6} = 0, L = \frac{a \cdot 3}{120 \cdot 2}$$

$$a = 6, b = 6, c = 0, L = \frac{3}{40}$$

2. Ans.(A,D)

Let $\sec x = t$

$$\therefore f(x) = [t] + \sqrt{t - [t]}$$

$$\text{As } 0 \leq x < \frac{\pi}{2}, 1 \leq t < \infty$$

$\Rightarrow$ Possible points of discontinuities may be at which $t \in \mathbb{N}$.

Let $k \in \mathbb{N}$

$$\text{LHL at } t = k, \lim_{t \rightarrow k^-} [t] + \sqrt{t - [t]}$$

$$= \lim_{h \rightarrow 0} (k-1) + \sqrt{k-h-k+1} = k$$

$$\text{RHL at } t = k, \lim_{t \rightarrow k^+} [t] + \sqrt{t - [t]}$$

$$= \lim_{h \rightarrow 0} k + \sqrt{k+h-k} = k$$

$\Rightarrow$ Function is continuous for all $x \in \left[0, \frac{\pi}{2}\right)$

3. Ans.(B,C)

$$f \circ g(0) = f(g(0)) = f(-3)$$

$$= \max\{1 - \sin 3, 3 - \cos 3, 2\} = 3 - \cos 3 > 2$$

$$g \circ f(0) = g(f(0)) = g(2) = -1$$

4. Ans.(B,C,D)

(A) Let $\sin^{-1} x = \theta \Rightarrow \sin \theta = x$

$$f(x) = \sin 2\theta = 2 \sin \theta \cos \theta = 2x \sqrt{1-x^2}$$

$$f(0) = 0 \Rightarrow \text{at } x = 0, 1, -1 \text{ (many-one)}$$

(B) $f(x) + f(-x) = 0 \Rightarrow$ odd function

(C) $g(x) = 2x$, continuous in given interval

(D) $f(x)$ is continuous function

5. Ans.(A,B,C)

$$L_1 \equiv x + 2y - 4 = 0$$

$$L_2 \equiv 3x - ky + 2 = 0$$

$$L_3 \equiv 5x + 3y + 1 = 0$$

If L_1, L_2 and L_3 are concurrent then

$$\left. \begin{aligned} x + 2y - 4 &= 0 \\ 5x + 3y + 1 &= 0 \end{aligned} \right\} \Rightarrow \begin{aligned} x &= -2 \\ y &= 3 \end{aligned}$$

$$3(-2) - k(3) + 2 = 0 \Rightarrow -\frac{4}{3} = k$$

$$\text{If } L_1, L_2 \text{ parallel } \frac{1}{3} = \frac{2}{-k} \neq \frac{-4}{2}$$

$$\Rightarrow k = -6$$

$$\text{If } L_2, L_3 \text{ parallel } \frac{3}{5} = -\frac{k}{3} \Rightarrow k = -\frac{9}{5}$$

6. Ans.(A,C)

$$f(x) = 2x^2 + kx + 50$$

$$|\alpha - \beta| = 8 \Rightarrow \frac{\sqrt{D}}{|a|} = 8 \Rightarrow D = 256$$

$$\therefore \text{Minimum value} = -\frac{D}{4a} = -\frac{256}{8} = -32$$

$$\therefore \text{at } x = 0, f(x) = 50$$

$\Rightarrow$ graph cuts positive y-axis

SECTION-II

1. Ans. 500.00

$$\text{Let } h(x) = f(x) - x = 2006(x-1)(x-2)(x-3) \dots (x-2005)$$

$$\text{Now } (f(2006) - 2006) = 2006.2005.2004 \dots 1$$

$$= 2006!$$

exponent of 5 in 2006!

$$= \left[\frac{2006}{5} \right] + \left[\frac{2006}{25} \right] + \left[\frac{2006}{125} \right] + \left[\frac{2006}{625} \right]$$

$$= 401 + 80 + 16 + 3 = 500$$

number of zeroes = 500

2. Ans. 13.00

$$\frac{\Delta}{s} = 2 \text{ and } \frac{\Delta}{s-a} = 15$$

$$\Rightarrow \frac{s-a}{s} = \frac{2}{15} \Rightarrow s = \frac{15}{13}a$$

$$\Rightarrow \Delta = \frac{30}{13}a = \frac{1}{2}bc$$

$$\Rightarrow bc = \frac{60}{13}a \quad \dots(1)$$

$$\text{Also } b + c = 2s - a = \frac{30}{13}a - a = \frac{17}{13}a \quad \dots(2)$$

$$\text{Now } b^2 + c^2 = a^2 \Rightarrow (b+c)^2 - 2bc = a^2$$

$$\Rightarrow \frac{289}{169}a^2 - \frac{120}{13}a = a^2 \text{ gives } a = 13$$

3. Ans. 0.06

$$\sin 5^\circ \cos 15^\circ \sin 55^\circ \sin 65^\circ$$

$$= \frac{1}{4} \sin 15^\circ \cos 15^\circ$$

$$= \frac{\sin 30^\circ}{8} = \frac{1}{16}$$

$$\therefore a = 1, b = 2 \Rightarrow a + b = 3$$

4. Ans. 4.00

$$|x| + y = 3 \text{ and } \left| \sin \frac{\pi x^2}{2} \right| = 1$$

$$\Rightarrow x = \pm 1 \text{ and } y = 2$$

$$\text{or } x = \pm 3 \text{ and } y = 0$$

$$\Rightarrow 4 \text{ ordered pairs}$$

5. Ans. 0.50

Equation of tangent at $(x_1, x_1 \sin x_1)$ is

$$y - x_1 \sin x_1 = (\sin x_1 + x_1 \cos x_1)(x - x_1)$$

$$\begin{aligned} \text{y intercept} &= x_1 \sin x_1 - x_1 \sin x_1 - x_1^2 \cos x_1 \\ &= -x_1^2 \cos x_1 \end{aligned}$$

x intercept

$$= x_1 - \frac{x_1 \sin x_1}{(\sin x_1 + x_1 \cos x_1)} = \frac{x_1^2 \cos x_1}{(\sin x_1 + x_1 \cos x_1)}$$

$$\text{sum} = \frac{x_1^2 \cos x_1}{(\sin x_1 + x_1 \cos x_1)} - x_1^2 \cos x_1$$

$$\mu(x_1) = \frac{x_1^2 \cos x_1 - x_1^2 \sin x_1 \cos x_1 - x_1^3 \cos^2 x_1}{(\sin x_1 + x_1 \cos x_1)}$$

$$\frac{\mu(x_1)}{x_1} = \frac{x_1^2 \cos x_1 - x_1^2 \sin x_1 \cos x_1 - x_1^3 \cos^2 x_1}{(x_1 \sin x_1 + x_1^2 \cos x_1)}$$

$$= \text{divide Nr \& Dr by } x_1^2$$

$$= \lim_{x_1 \rightarrow 0} \frac{\mu(x_1)}{x_1} = \frac{1}{2}$$

6. Ans. 8.00

$$\text{at } x = 0$$

$$\text{at } x = -1$$

$$\begin{aligned} \text{LHL} &= b & f(-1) &= 5 + c \\ f(0) &= 2 & f(-1^+) &= b + 2 \end{aligned}$$

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0} \ell n (\cos x + a \sin bx)^{1/x} & f(-1^-) &= 5 + c \\ &= ab & \text{so } 5 + c &= b + 2 \\ \text{so } ab &= 2 = b & c &= -1 \\ b &= 2, a = 1 \\ \cos^{-1} \cos(a + c + 3b) &= \cos^{-1} \cos 6 = 2\pi - 6 \end{aligned}$$

SECTION-III

1. **Ans. 1**

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(\sin x)^2}{(e^{\tan^2 x} - 1)} \\ = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \times \frac{\tan^2 x}{(e^{\tan^2 x} - 1)} \times \left(\frac{x}{\tan x} \right)^2 = 1 \end{aligned}$$

2. **Ans. 1**

$$\begin{aligned} f(0) &= b \\ \lim_{h \rightarrow 0^+} f(0-h) &= \lim_{h \rightarrow 0^+} \frac{\sin(a+2)h + \sinh}{h} = a + 3 \\ \lim_{h \rightarrow 0^+} f(0+h) &= \lim_{h \rightarrow 0^+} \frac{(3h^2 + h)^{1/3} - (h)^{1/3}}{(h)^{4/3}} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{3h^2 + h - h}{(h)^{4/3} ((3h^2 + h)^{2/3} + (h)^{2/3} + (3h^2 + h)^{1/3} (h)^{1/3})} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{3h^2}{h^2 ((3h+1)^{2/3} + 1 + (3h+1)^{1/3})} = 1 \\ b &= 1, a = -2 \\ |a + b| &= 1 \end{aligned}$$

3. **Ans. 3**

$$\sum_{r=1}^n \tan^{-1} \left(\frac{2}{(r+1)^2} \right) = \tan^{-1} \left(\frac{(r+2)-r}{1+r(r+2)} \right)$$

$$\begin{aligned} &= (\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 4 - \tan^{-1} 2) + \\ &(\tan^{-1} 5 - \tan^{-1} 3) + \dots (\tan^{-1} (n+1) - \tan^{-1} (n-1)) \\ &+ (\tan^{-1} (n+2) - \tan^{-1} n) \\ &= \tan^{-1} (n+1) + \tan^{-1} (n+2) - \tan^{-1} 1 - \tan^{-1} 2 \\ \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2}{(r+1)^2} \right) &= \frac{\pi}{2} + \frac{\pi}{2} - \tan^{-1} 1 - \tan^{-1} 2 \\ &= \tan^{-1} 3 \end{aligned}$$

4. **Ans. 4**

$$f(x) = \tan^{-1} \sqrt{[x] + [-x]} + \sqrt{2-|x|} + \frac{1}{x^2}$$

$$[x] + [-x] = \begin{cases} -1 & x \notin I \\ 0 & x \in I \end{cases}$$

$$\begin{aligned} \text{Also, } [x] + [-x] &\geq 0 \\ \Rightarrow x &\in I \end{aligned}$$

$$\text{Also, } 2 - |x| \geq 0$$

$$|x| \leq 2$$

$$-2 \leq x \leq 2$$

$$x \neq 0$$

$$\Rightarrow x \in \{-2, -1, 1, 2\}$$

5. **Ans. 1**

$$\begin{vmatrix} 1 & p & 2 \\ 2 & 1 & 2 \\ 1 & 1 & 2 \end{vmatrix} = 0 \Rightarrow p = 1$$

6. **Ans. 2**

$$\begin{aligned} (b^4 + a^2c^2 - 2ab^2c) + (c^4 + b^2d^2 - 2bc^2d) \\ + (a^2d^2 + b^2c^2 - 2abcd) \leq 0 \\ (b^2 - ac)^2 + (c^2 - bd)^2 + (ad - bc)^2 \leq 0 \end{aligned}$$

$$\Rightarrow b^2 = ac, c^2 = bd \text{ \& } \frac{b}{a} = \frac{d}{c}$$

$$\Rightarrow a, b, c, d \text{ in G.P.}$$

$$\text{but } a, b, c, d \text{ are in A.P.}$$

$$\Rightarrow a = b = c = d.$$